# ACTRIS ACSM Near Real Time submissions manual

## Software suite installation

### 1. IGOR ACSM Export procedure

One must install the src/igor/ACMCC_Export.ipf IGOR procedure. It will generate raw text files directly from the acquisition software.

Installation and setup is detailled in the ACSM_Export_Tool_Manual.pdf document.

### 2. ACTRIS ACSM processing software

### 3. Python environment

1. Install python3
    a. Download the python installer at https://www.python.org/downloads
        Note that if you are on a windows 7 platform, the latest compatible version is 3.8
    b. Right click on downloaded file, then “execute as an administrator”
    c. Click on the select box “Add Python 3.x to PATH”
    d. Click on “Install now”

2. Install required dependencies
    a. double-click on the install_python_packages.bat file
    Note that depending of your platform, some packages could be missing, but the software is still functional.

### 4. ACSM NRT Processing software

1. Install the ACTRIS-ACSM-CONVERTER package
    a. Download the latest release at https://www.icare.univ-lille.fr/depot/ACTRIS/ACMCC/software/actris_acsm_converter-latest.zip
    b. Unzip it where you want to install it. Recommended is in your home directory. As an example, to C:\Users\pascal\actris_acsm_converter. This is what will be assumed in the rest of this manual

2. Configure the raw -> lev0 converter for your product
    a. Configure your product infos, in cfg/product.cfg, add an entry like

```
[GIF_ACSM_140113_RAW_NRT]
# enable transfer of this product if set to 1
enable=1
instrument_sid = GIF_ACSM_140113
platform_sid = GIF
description = Quadrupole ACSM #140113 - SIRTA station - Native format
regex = SIRTA_ACSM-140113_(?P<year>[0-9]{4})(?P<month>[0-9]{2})(?P<day>[0-9]{2})(?
P<hour>[0-9]{2})(?P<min>[0-9]{2}).txt
```

Add your own product ID (in green) and set an ID for your instrument (in blue). If your platform is not yet known, also add an ID for your station (in orange).

The description is free, and the most important part is to set the regular expression that match your raw files. It will be used to identify the process to be ran.

For the different product IDs, it is recommended to follow the conventions:

- <station-ID>: the GAW-ID of your station, if existing, else set one 3 characters
- <instrument-ID>: <station-ID>_ACSM_<instrument-serial-number>
- <product-ID>: <instrument-ID>_RAW_NRT

b. Prepare your actris header template, in cfg/actris_header directory, copy one existing to a file named <product-id>.actris_header

- The file prefix title (in green) must match the one you set in the product.cfg file,
- the entries like ${...} must not be changed, they are dynamically be filled at runtime.
- For the rest, change it for your instrument configuration.

c. Configure your instrument infos, in cfg/instrument.cfg, add an entry like

```
[GIF_ACSM_140113]
model=Q-ACSM
# serial number, make sure to add quotes, else interpreted as an operation
sn = "140-113"
# PI contact ID, must be in contact.cfg file
contact_sid=JeP
```

- The section title (in blue) must match the one you set in the product.cfg file,
- model: can be Q-ACSM or ToF-ACSM,
- sn: the serial number (with double quotes),
- contact_sid: the instrument PI ID,

d. **If not already defined**, configure the PI infos, in cfg/contact.cfg, add an entry like

```
[JeP]
first_name=Jean-Eudes
last_name=Petit
email=jean-eudes.petit@lsce.ipsl.fr
institution_sid=LSCE-CAE
```

- The section title (in purple) must match the one you set in the instrument.cfg file,
- the rest is self-descriptive I think

e. **If not already defined**, configure the station infos, in cfg/platform.cfg, add an entry like

```
[GIF]
name=Gif Sur Yvette (91)
latitude=48.709
longitude=2.159
altitude=162
contact_sid=VG
# --- EBAS related metadata
station = FR0020R
platform = FR0020S
lab = FR01L
wmo_land_use = Agricultural
```

```
wmo_setting = Suburban
gaw_type = C
wmo_region = 6
gaw_id = GIF
site = SIRTA
station_name = SIRTA
station_gaw_name = SIRTA Atmospheric Research Observatory
wdca_id = GAWAFR__GIF
```

- The section title (in orange) must match the one you set in the platform.cfg file,
- The rest is mostly the information requested in the EBAS templates

f. **If not already defined**, configure the EBAS upload infos, in cfg/upload.cfg, add an entry like

```
[GIF]
user=FR0020
```

- The section title (in orange) must match the one you set in the platform.cfg file,
- the user is the FTP push username defined for you by EBAS

3. Create a ssh key (if not already set)
   a. Download and install git for windows, with the default configuration (a lot of confirm clicks): https://git-scm.com/download/win
   b. Click on the windows key, type “git bash”, under best match, click “git bash”
   c. In the command prompt, type “ssh-keygen –t rsa”
   d. Press “enter” 3 times to use the default
   e. 2 files have been created under C:\Users\<username>\.ssh Your private key is C:\Users\<username>\.ssh\id_rsa and your public one is C:\Users\<username>\.ssh\id_rsa.pub
   f. Send an email with your public key C:\Users\<username>\.ssh\id_rsa.pub to ebas@nilu.no and ask that this key to be added to the sftpush authorized_keys for your account. Don’t forget to specify your login.

4. Set the format/upload process as a scheduled task
   a. We assume that your ACMCC_Export directory is in your home: C:\Users\pascal\ACMCC_Export
   b. Create the directory where will be written the lev0 files. By default, C:\Users\pascal\ACMCC_Export\lev0 is used
   c. Set it as a scheduled task
      i. Press the windows touch + R, and then enter “taskschd.msc”. It opens the windows task scheduler.
      ii. Create a new task:
         1. In the “general” panel
            a. name “ACMCC lev0 uploader”
            b. Click on “execute even if user not connected”
         2. In the Trigger panel
            a. Click on New
            b. In the Start entry, set the hour to “<current-hour>:30:00
            c. Select “repeat each hour”, since “infinitely”
         3. In the “Actions” panel,

   a. Select “New”
   b. On the “program/script” line, select “Browse”
   c. Select the “run_uploader.bat” script at the root of the package (eg C:\Users\pascal\actris_acsm_converter\run_uploader.bat)
   d. In “Start into", enter the directory that contains the script “C:\Users\pascal\actris_acsm_converter”
4. Select OK, you eventually need to enter the admin password at this point

## 5. ACSM Regular Annual Submission software

One must install the src/igor/ACSM_Regular_Submission_Tool.ipf IGOR procedure. It will generate raw text files directly from the acquisition software.

Installation and setup is detailled in the doc/ACSM_Regular_Submission_Tool_Manual.pdf document.